

it just shows the IP address or URL. Use the following code modification, reload the browser, and check the tab and source code again:

```
#!/usr/local/bin/perl

print "Content-type: text/html\n\n";
print "<html>\n <head>\n  <title>My Cloud Magic with Perl</title>\n </head>\n";
print " <body>\n  ";
foreach $var (sort(keys(%ENV))) {
    print "$var=".ENV{$var}."<br>\n  ";
}
print "</body>\n </html>";
```

I am not referring to any *CGI.pm* functions. We are trying to use the easiest methods to write HTML code. Unless it is to pass parameters from one script/HTML page to the other, the *CGI.pm* module is very cumbersome to do effective/fast HTML coding.

Table format

We will convert the *key=value* pair format to a table, for which we can use the following code:

```
#!/usr/local/bin/perl

print "Content-type: text/html\n\n";
print "<html>\n <head>\n  <title>My Cloud Magic with Perl</title>\n </head>\n";
print " <body>\n  <table>\n    ";
foreach $var (sort(keys(%ENV))) {
    print "<tr><td>$var</td><td>".ENV{$var}."</td></tr>\n    ";
}
print "</table>\n </body>\n</html>";
```

Style sheet

We will apply a six-line style sheet to our *printenv*, as follows:

```
#!/usr/local/bin/perl

print "Content-type: text/html\n\n";
print "<html>
<head>
  <title>My Cloud Magic with Perl</title>
  <style>
    tr:nth-child(even) {background: #FFC300}
    tr:nth-child(odd)  {background: #FF8B00}
    td:nth-child(1)    {font-weight: bold; font-family:
'Courier New', Times, serif;}
    td:nth-child(2)    {font-style: italic; width: 500px}
  </style>
</head>
```

```
<body>
  <table><tr><th>Environment Variable</th><th>Value</th></tr>
  ";
  foreach $var (sort(keys(%ENV))) {
    print "    <tr><td>$var</td><td>".ENV{$var}."</td></tr>\n";
  }
  print "  </table>\n </body>\n</html>";
```

Now try out both URLs and their source code, as well. We can be very proud of ourselves for getting such a raw output to look so good!

You can change your background colour to suit your taste using <https://htmlcolorcodes.com/>, and the font family, style and weight using https://www.w3schools.com/css/css_font.asp.

We are pretty much done with looking at what *mod_cgi* and *mod_perl* have to offer. There are more books and tutorials out there on this topic if you want to explore the subject in more depth.

At <http://www.cgi101.com/book/ch3/text.html> you will get more details on *CGI.pm* module usage, and you can compare the syntax we used in this article, look at how it can be equally represented with the *CGI.pm* module, and what most of these environment variables stand for.

The next article will be on templates, in which we will separate style sheets from the main page and use a bit of JavaScript as well. Till then, sail happily while using Perl to hack on the cloud.

Errors

If you see any 500 HTTP server errors as you run the code, please do not forget to check the Apache logs. This is very helpful in troubleshooting. In my experience, Perl is friendlier than Python, when it comes to debugging messages. 

References

- [1] 'CGI Programming with Perl' by Scott Guelich, Shishir Gundavaram and Gunther Birznies
- [2] 'Practical mod_perl: Programming, Administration, Performance Tips' by Eric Cholet and Stas Bekman.
- [3] https://www.w3.org/Protocols/rfc1341/7_1_Text.html
- [4] https://www.tutorialspoint.com/apache_bench/
- [5] <https://bugzilla.readthedocs.io/en/latest/installing/apache.html>
- [6] <https://opensourceforu.com/2015/11/scripting-mail-with-shell-and-perl/>
- [7] https://httpd.apache.org/docs/2.4/mod/mod_env.html
- [8] <http://blog.builtinperl.com/post/5-major-websites-that-use-perl>

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